

Here is the **complete, consolidated list of all operations and tests** for the EdTech Study Platform (SRS v7.3), incorporating the original requirements, the strict **Supabase + Backblaze B2** serverless architecture, and the specific integrations for **Resend**, **VAPID**, and **Sentry**.

This list is organized by module for easy execution.

1. Authentication & Registration

• User Registration

- Register with valid Full Name, valid Indian Mobile (+91), valid Email, Strong Password.
- Register with invalid Mobile (non-Indian, wrong length, non-numeric).
- Register with weak password (missing uppercase/number/special char).
- Register with mismatched "Confirm Password".
- Register with duplicate email.
- Register with empty required fields.
- **Race Condition:** Simultaneous registration of two users (verify only first becomes super_admin).

• Email Verification (Resend Integration)

- Click valid verification link immediately.
- Click expired link (>24h).
- Click invalid/tampered link.
- Attempt login *before* verification (expect 403).
- Click "Resend Verification Email" (verify rate limit: max 3/hr).
- Verify email delivery to Gmail, Outlook, Yahoo (check Spam).
- Verify email content (Subject, Body, Link).

• Login & Session

- Login with valid credentials.
- Login with incorrect password.

- Login with unverified email.
- Session persistence (refresh page).
- Session expiration (JWT expiry).
- Logout and redirection.
- **Password Management**
 - Change password (correct current, new strong).
 - Change password (incorrect current).
 - Change password (same as old).
 - Change password (weak new).
 - Verify logout of other sessions (if implemented).
 - "Forgot Password" flow (request, reset, verify).

2. User Profile & Settings

- **Profile Photo (Backblaze B2)**
 - Upload valid JPG/PNG/WEBP (<2MB).
 - Upload >2MB (verify rejection).
 - Upload non-image (verify rejection).
 - Verify client-side compression/resizing.
 - Change photo (verify old file deletion from B2).
 - Remove photo (verify revert to default + B2 deletion).
 - Verify RLS: User A cannot see User B's photo.
- **Profile Editing**
 - Edit Full Name (valid).
 - Edit Mobile (valid Indian format).
 - Attempt to edit Email (verify read-only).
 - Save and verify DB update.
- **Preferences**
 - Toggle Dark/Light mode.
 - Update notification preferences.
 - Verify critical alerts cannot be disabled.

3. Content Hierarchy & Smart Gating

- **Navigation**
 - Navigate Subject → Chapter → Topic.

- Verify states: Locked (🔒), Unlocked (🔓), Completed (✓).
- Direct URL access to locked content (expect 403).
- Verify "Next Action" card logic.
- **Gate Logic Execution**
 - **Lecture:** Click "Watch" (verify Telegram deep link).
 - **Lecture Quiz:** Pass, Fail, Retry (limit), Exhaust attempts (block).
 - **DPP:** Attempt before Quiz (block), Complete (pass/fail).
 - **PYQs:** Attempt before DPP (block), Complete.
 - **Topic Test:** Attempt before PYQs (block), Pass.
 - **Chapter Test:** Attempt before all Topics (block), Pass (unlock PDF).
 - **Subject/Multi-Subject:** Verify unlock conditions.
- **Admin Configuration**
 - Change passing score/thresholds.
 - Verify immediate effect on new attempts.

4. Exam System

- **Start & Interface**
 - Start exam (verify server-time sync).
 - Verify Question Grid, Keyboard Shortcuts, Mobile Swipes.
- **During Exam**
 - Select answer, Mark for Review, Navigate Grid.
 - Pause/Resume (verify timer state).
 - Tab switch warning.
 - Copy/Paste block.
 - Network disconnect (verify IndexedDB save).
 - Browser crash (verify "Resume" prompt).
 - Prolonged disconnect (>10m) → Auto-submit.
- **Submission**
 - Manual submit, Timer expiry auto-submit.
 - Partial answers, All marked.
 - Verify negative marking and score calculation (precision).
- **Post-Exam**
 - View Dashboard (Score, Accuracy, Percentile).

- View Answer Sheet (User vs. Correct).
- View Text Solutions (LaTeX rendering).
- Scan QR Code (verify YouTube redirect).
- Attempt Answer Sheet before submission (block).
- Filter "Incorrect Only" / "Skipped Only".

5. Storage & Files (Backblaze B2)

- **PDF Upload (Gated)**
 - Upload to unlocked Chapter.
 - Upload to locked Chapter (403).
 - Upload >10MB (reject).
 - Upload non-PDF (reject).
 - User quota full (500MB) → 413/501.
 - Global quota full (9GB) → 429.
 - Duplicate file (hash check).
- **Retrieval & Search**
 - Download/View PDF (verify Signed URL).
 - Search PDF text (verify index).
 - Verify CDN caching (headers, speed).
 - Delete note (verify B2 delete + CDN purge).
- **Photo Storage**
 - Verify B2 path structure (users/{uid}/profile/).
 - Verify versioning (if enabled).

6. Productivity Tools

- **Pomodoro Timer**
 - Start Standard/Long/Custom.
 - Pause/Resume/Abort.
 - Complete session (log + chime).
 - Background execution (Service Worker).
 - Offline mode (local sync).
 - Verify "Focus Streak" logic.
- **Smart Study Planner**
 - View auto-tasks (weak topics, spaced rep).

- Create/Edit/Delete manual tasks.
- Update status (Pending → Done).
- Drag & Drop reorder.
- Verify auto-tasks respect gating (no locked content).
- **External Test Logging**
 - Log test (Name, Date, Score, Rank).
 - Edit/Delete entry.
 - Verify chart update.
 - Invalid data (Score > Total) → Validation error.

7. Admin & Super Admin

- **User Management**
 - View pending queue.
 - Approve/Reject users.
 - Suspend/Ban users.
 - Promote/Demote roles.
 - Verify "Sole Super Admin" cannot demote self.
 - Reset user progress.
- **Content Management**
 - CRUD Subjects/Chapters/Topics.
 - Bulk import Questions (CSV/JSON) + Validation.
 - Configure Telegram links.
 - Create/Edit Exams (DPP/Test).
 - Set negative marking.
 - Upload Text Solutions (Markdown/LaTeX).
 - Paste YouTube URL (auto-QR).
 - Map video to Question/Topic.
 - Publish/Unpublish Answer Sheets.
- **System Config**
 - Modify system_config (limits, scores).
 - View Audit Logs.
 - Trigger manual storage alert.

8. Security & Edge Cases

- **Access Control (RLS)**
 - User A view User B's notes (block).
 - User A view User B's attempts (block).
 - Unapproved user access (block).
 - Admin view personal data (block).
- **Rate Limiting**
 - Rapid quiz attempts (429).
 - Rapid PDF uploads (429).
 - Rapid registration (429).
 - Rapid password changes (429).
- **Data Integrity**
 - SQL Injection attempts.
 - XSS attempts.
 - Foreign key constraints (cascade).
 - Transaction atomicity.
- **Network & Resilience**
 - 3G latency (load <2s).
 - Packet loss (retry logic).
 - B2 API failure (graceful error).
 - Supabase Auth failure.
 - Offline exam sync queue.
- **Time & Sync**
 - Client clock manipulation (server override).
 - Timezone conversion.
 - UTC storage vs. Local display.

9. Performance & Load

- **Concurrency**
 - 20, 500, 2,000 concurrent users.
 - 3x peak traffic spike.
- **Latency**
 - Gate Check (<50ms).
 - Exam Start.
 - B2 Signed URL generation.
 - FCP on 3G.

- **Storage**
 - Simultaneous upload speed.
 - CDN cache hit ratio.
 - B2 egress reduction.

10. Compatibility & Usability

- **Browsers:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+, Incognito.
- **Devices:** Mobile (320px-768px), Tablet, Desktop, Landscape/Portrait.
- **Accessibility:** Keyboard nav, Screen readers, Contrast, Focus indicators.
- **Localization:** English UI rendering.

11. External Integrations (Resend, VAPID, Sentry)

- **Resend (Email)**
 - **Security:** Verify API Key in Edge Secrets (not client).
 - **Triggers:** Verification, Welcome, Approval, Rejection, Storage Alert, Password Reset.
 - **Rate Limit:** 4 rapid resends → 429.
 - **Deliverability:** Test Gmail, Outlook, Yahoo, Spam folders.
 - **Error Handling:** Simulate Resend 500 → Graceful user message.
- **VAPID (Web Push)**
 - **Security:** Verify Private Key in Secrets, Public Key in client.
 - **Permissions:** Allow/Block prompts.
 - **Triggers:** Daily Plan, Streak, Exam Reminder, New Task.
 - **Payload:** Click notification → Correct redirect.
 - **Preferences:** Toggle off specific types (verify suppression).
 - **Critical Alerts:** Verify "Account Suspended" bypasses user toggles.
 - **Cleanup:** Uninstall PWA → Verify subscription cleanup.
- **Sentry (Monitoring)**
 - **Security:** Verify DSN exposure (safe) vs. Auth Token (hidden).

- **Capture:** Client-side JS error, Edge Function error, Network timeout.
- **Context:** Verify User ID, Trace ID, Breadcrumbs.
- **Alerts:** Simulate >5% error rate → Slack/Email alert.
- **Privacy:** Verify PII (passwords, mobile) is scrubbed.
- **Integrated Security**
 - **Secret Leakage:** Inspect build bundle (no API keys).
 - **Proxy Test:** Direct API call from browser (fail).
 - **Token Expiry:** Resend link (24h), B2 URL (15m).
 - **Rate Limit Propagation:** Resend 429 handled by Edge Function.

12. Data Retention & Archival

- **Archival:** Move user_attempts >12 months to archive table.
- **Backup:** Verify Supabase daily backup.
- **Restore:** Simulate restore (verify integrity).
- **B2 Versioning:** Verify retention of last 3 versions.
- **GDPR:** Export all user data (JSON).

13. Deployment & CI/CD

- **Pipeline:** PR tests, Vercel deploy, Edge Function deploy, DB migrations.
- **Rollback:** Auto-rollback on failure, Manual rollback.
- **Secrets:** Rotation of Resend/B2 keys (zero downtime).
- **Env Vars:** Verify staging vs production separation.

This comprehensive list ensures every aspect of the **SRS v7.3** is validated, from the smallest input field to the complex serverless architecture and third-party integrations.